

Supplementary Materials: On Calibration of Modern Neural Networks

Dataset	Model	Uncalibrated	Hist. Binning	Isotonic	BBQ	Temp. Scaling	Vector Scaling	Matrix Scaling
Birds	ResNet 50	22.54%	55.02%	23.37%	37.76%	22.54%	22.99%	29.51%
Cars	ResNet 50	14.28%	16.24%	14.9%	19.25%	14.28%	14.15%	17.98%
CIFAR-10	ResNet 110	6.21%	6.45%	6.36%	6.25%	6.21%	6.37%	6.42%
CIFAR-10	ResNet 110 (SD)	5.64%	5.59%	5.62%	5.55%	5.64%	5.62%	5.69%
CIFAR-10	Wide ResNet 32	6.96%	7.3%	7.01%	7.35%	6.96%	7.1%	7.27%
CIFAR-10	DenseNet 40	5.91%	6.12%	5.96%	6.0%	5.91%	5.96%	6.0%
CIFAR-10	LeNet 5	15.57%	15.63%	15.69%	15.64%	15.57%	15.53%	15.81%
CIFAR-100	ResNet 110	27.83%	34.78%	28.41%	28.56%	27.83%	27.82%	38.77%
CIFAR-100	ResNet 110 (SD)	24.91%	33.78%	25.42%	25.17%	24.91%	24.99%	35.09%
CIFAR-100	Wide ResNet 32	28.0%	34.29%	28.61%	29.08%	28.0%	28.45%	37.4%
CIFAR-100	DenseNet 40	26.45%	34.78%	26.73%	26.4%	26.45%	26.25%	36.14%
CIFAR-100	LeNet 5	44.92%	54.06%	45.77%	46.82%	44.92%	45.53%	52.44%
ImageNet	DenseNet 161	22.57%	48.32%	23.2%	47.58%	22.57%	22.54%	-
ImageNet	ResNet 152	22.31%	48.1%	22.94%	47.6%	22.31%	22.56%	-
SVHN	ResNet 152 (SD)	1.98%	2.06%	2.04%	2.04%	1.98%	2.0%	2.08%
20 News	DAN 3	20.06%	25.12%	20.29%	20.81%	20.06%	19.89%	22.0%
Reuters	DAN 3	2.97%	7.81%	3.52%	3.93%	2.97%	2.83%	3.52%
SST Binary	TreeLSTM	11.81%	12.08%	11.75%	11.26%	11.81%	11.81%	11.81%
SST Fine Grained	TreeLSTM	49.5%	49.91%	48.55%	49.86%	49.5%	49.77%	48.51%

Table S2. Test error (%) on standard vision and NLP datasets before calibration and with various calibration methods. The number following a model’s name denotes the network depth. Error with temperature scaling is exactly the same as uncalibrated.

Dataset	Model	Uncalibrated	Hist. Binning	Isotonic	BBQ	Temp. Scaling	Vector Scaling	Matrix Scaling
Birds	ResNet 50	0.9786	1.6226	1.4128	1.2539	0.8792	0.9021	2.334
Cars	ResNet 50	0.5488	0.7977	0.8793	0.6986	0.5311	0.5299	1.0206
CIFAR-10	ResNet 110	0.3285	0.2532	0.2237	0.263	0.2102	0.2088	0.2048
CIFAR-10	ResNet 110 (SD)	0.2959	0.2027	0.1867	0.2159	0.1718	0.1709	0.1766
CIFAR-10	Wide ResNet 32	0.3293	0.2778	0.2428	0.2774	0.2283	0.2275	0.2229
CIFAR-10	DenseNet 40	0.2228	0.212	0.1969	0.2087	0.1750	0.1757	0.176
CIFAR-10	LeNet 5	0.4688	0.529	0.4757	0.4984	0.459	0.4568	0.4607
CIFAR-100	ResNet 110	1.4978	1.4379	1.207	1.5466	1.0442	1.0485	2.5637
CIFAR-100	ResNet 110 (SD)	1.1157	1.1985	1.0317	1.1982	0.8613	0.8655	1.8182
CIFAR-100	Wide ResNet 32	1.3434	1.4499	1.2086	1.459	1.0565	1.0648	2.5507
CIFAR-100	DenseNet 40	1.0134	1.2156	1.0615	1.1572	0.9026	0.9011	1.9639
CIFAR-100	LeNet 5	1.6639	2.2574	1.8173	1.9893	1.6560	1.6648	2.1405
ImageNet	DenseNet 161	0.9338	1.4716	1.1912	1.4272	0.8885	0.8879	-
ImageNet	ResNet 152	0.8961	1.4507	1.1859	1.3987	0.8657	0.8742	-
SVHN	ResNet 152 (SD)	0.0842	0.1137	0.095	0.1062	0.0821	0.0844	0.0924
20 News	DAN 3	0.7949	1.0499	0.8968	0.9519	0.7387	0.7296	0.9089
Reuters	DAN 3	0.102	0.2403	0.1475	0.1167	0.0994	0.0990	0.1491
SST Binary	TreeLSTM	0.3367	0.2842	0.2908	0.2778	0.2739	0.2739	0.2739
SST Fine Grained	TreeLSTM	1.1475	1.1717	1.1661	1.149	1.1168	1.1085	1.1112

Table S3. NLL (%) on standard vision and NLP datasets before calibration and with various calibration methods. The number following a model’s name denotes the network depth. To summarize, NLL roughly follows the trends of ECE.